Seminar 1

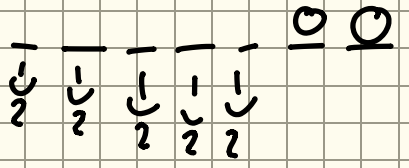
1)

A B C D

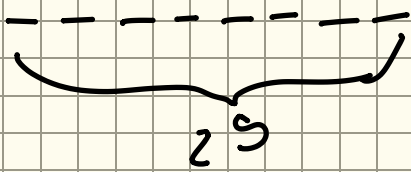
$\neq 0, \neq D$ | $\neq A, \neq D$ | $\neq B, \neq D$ | $\{1, 3, 5, 7, 9\}$ $\Rightarrow 8 \cdot 8 \cdot 7 \cdot 5 = 64 \cdot 35 = 2240$

8 options · 8 options · 7 options · 5 options

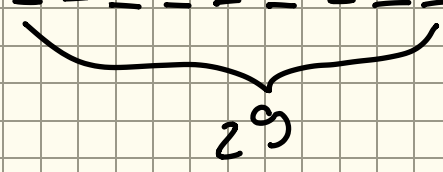
2) a) $2^4 = 3$



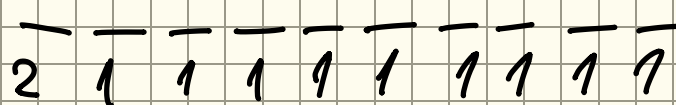
b) $2^9 = 512$

c)  $\Rightarrow$ 1 if odd, 0 if even

2^9

d)  $\Rightarrow$ 1 if even, 0 if odd

2^9

e)  $\Rightarrow$ 2 ways

2 1 1 1 1 1 1 1 1 1

L1, Ex 6: A B C D E F G H

$\{C, H\} = \{H, C\}$

6) $A = \{1, 2, \dots, 10\}$

a) $\{3\} \rightarrow \{4, 5, \dots, 10\}$ $2^7 - 1 + 1 = 2^6 = 128$

b) 1 2 3 4 5 6 7 8 9 10

9 7 5 3 1

At each step, we choose the opponent of the smallest number with that opponent.

$\{1,3\}$ 9 ↑ 5 $9 \cdot 7 \cdot 5 \cdot 3 = 9!!$

$\{2,5\}$ (2n people → n pairs) $= (2n-1)!!$

$\{4,7\}$

$\{6,8\}$

$\{9,10\}$

8) $4 = 1+0+0+0$

$4 = 3+1+0+0$

$4 = 2+1+1+0$

$4 = 2+2$

$4 = 1+3$

$4 = 1+2+1+0$

$4 = 1+1+2+0$

$4 = 1+1+1+1$